2 Structure of Atom

SOLUTION (QR CODE)

2. (i) Mass of one electron = 9.10939×10^{-31} Kg.

∴ Number of electrons that weigh 9.10939×10^{-31} Kg = 1

Number of electrons that will weigh 1 g

= (1×10^{-3} Kg.)

$$= \frac{1}{9.10939 \times 10^{-31} kg} \times \left(1 \times 10^{-3} kg\right)$$

= $0.1098 \times 10^{28} = 1.098 \times 10^{27}$

(thoda sa dimag lgana hai bas)

(ii) Mass of one electron = 9.10939×10^{-31}Kg.

Mass of one mole of electron
= $(6.022 \times 10^{23}) \times (9.10939 \times 10^{-31}$Kg)

= 5.48×10^{-7}kg.

Charge on one electron

= 1.6022×10^{-19} coulomb

Charge on one mole of electron
= $(1.6022 \times 10^{-19})(6.022 \times 10^{23})$

= 9.65×10^{4}C

11. Convert the kinetic energy from eV to Joules

1 eV = 1.602×10^{-19} J, so,

20 eV = $20 \times 1.602 \times 10^{-19}$ J = 3.204×10^{-18} J

Convert the mass from grams to kilograms

m = 1.67×10^{-23}g = 1.67×10^{-26}kg

$K.E = \frac{1}{2} mv^2$

Rearrange the above equation for v :

$v = \sqrt{(2KE / m)}$

$$= \sqrt{\frac{\left(2 \times 3.204 \times 10^{-18} J\right)}{1.67 \times 10^{-26} kg}} = \approx 1.96 \times 10^{4} \text{ m/s}$$

baap re baap, pagal ho gaye kya

13. KE = $(1/2)mv^2$, eV = KE

V = KE/e

$$= \left(\frac{1}{2}\right)\frac{mv^2}{e}$$

$$= \left(\frac{1}{2}\right) \times 9.109 \times 10^{-31} kg \times \frac{(8.38 \times 10^{6} m/s)^2}{1.602 \times 10^{-19} C}$$

= 20 V

pehle sawaal ko toh samjho kehna kya chahta hai

19. 194×10^{3} J/mole ⇒ to break one mol of Br –Br bond , the required energy
= 194×10^{3} J

Energy required to break one Br – Br molecule

$$\Rightarrow \frac{194 \times 10^{3}}{6.02 \times 10^{23}} J / \text{molecule} = \frac{hc}{\lambda}, \frac{1}{\lambda}$$

= 1.62×10^{6} m^{-1}

23. $E = \frac{hc}{\lambda}$

$$= \left(\frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{620 \times 10^{-9}}\right) = 3.20 \times 10^{-19} \text{ J}$$

No. of photons emitted per second is:

$$= \frac{\left\{5 \times 10^{-3} J\right\}}{\left\{3.14 \times 10^{-19}\right\} J} = 1.6 \times 10^{16}$$

agr yeh nhi kr paye toh tum lallu ho

31. (A) $\frac{1}{2}mv^2 = K.E$, mv = p

$v = \frac{p}{m}$

$KE = \frac{1}{2} m \left(\frac{p^2}{m^2}\right)$

$K.E = \frac{p^2}{2m}$

hν = W + KE

$$\frac{hc}{\lambda} = W + K \cdot E = W + \frac{p^2}{2m} \simeq \frac{p^2}{2m}$$

$$\frac{hc}{\lambda'} = W + \frac{(1.5p)^2}{2m} \simeq \frac{9p^2}{8m}$$

$$\frac{\frac{hc}{\lambda} = \frac{p^2}{2m}}{\frac{hc}{\lambda'} = \frac{9p^2}{8m}} = \lambda' = \frac{4}{9}\lambda$$

badhiya sawaal hai, isko dimag mei baithalo

36. (D) Rutherford's experiment, which established the nuclear model of the atom, used a beam of helium nuclei (α-particles), which impinged on a gold foil (of thickness 100 nm) and got scattered. The gold foil was surrounded by a ZnS fluorescent screen. Wherever alpha particles strike the screen, a tiny flash of light is produced at that point. α-particles are called a helium nucleus.

agar upar padha hai toh hi kar paoge

39. (C) Rutherford's experiment on scattering of α-particles showed for the first time that the atom has a nucleus. He observed that the positively charged α-particles were repelled and deflected by the positive charges in the atom. Rutherford named this positively charged of atom as nucleus.

yaad karna hi padega or koi chaara nahi hai

46. Calculate change in energy as:

$E_1 = 2.17 \times 10^{-11}$ erg

$$\Delta E = 2.17 \times 10^{-11}\left(1 - \frac{1}{2^2}\right) \times 10^{-7} J$$

$= 1.6275 \times 10^{-18}$ J

Now, calculate wavelength by using formula:

$\lambda = hc/\Delta E$

$$= \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{1.6275 \times 10^{-18}} m$$

$$= 122 \times 10^{-9} m = 1220 \text{ Å}$$

47. (A) As, $r = a_0 \frac{n^2}{Z}$

For Helium ion: Z = 2 and n = 2 (given)

$$\therefore r = a_0 \cdot \frac{(2)^2}{2} = 2a_0$$

53. (A) Energy of ground state

$He^+ = -13.6 \times 4$ eV $= -54.4 \times 4$ eV

(B) P.E of 1st orbit of H-atom

$= 2T.E. = -2 \times 13.6$ eV $= -27.2$ eV

(C) Energy of II excited state

$$= -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{(2)^2}{(3)^2}$$

$$= -13.6 \times \frac{4}{9} = -6.04 \text{ eV}$$

(D) I.E. $= -E_1 = 21.8 \times 10^{-19} \times 4J = 8.7 \times 10^{-18}$ J

aankh band karke formula socho bas

71. (C) $(He^+)_{2 \to 4} = (Li^{2+})_{n_4 \to n_3}$

$$\therefore \frac{Z_1}{Z_2} = \frac{n_2}{n_4} = \frac{n_1}{n_3}$$

or $\frac{2}{3} = \frac{2}{n_4} = \frac{4}{n_3}$

$\therefore$ $n_4 = 3$ and $n_3 = 6$

So, the transition in Li^{2+} ion will be from 3 to 6.

78. (B) The formula used for the determination of λ is

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_g^2}\right)$$

where, n_i is initial orbit n_g is final orbit

For H: $\frac{1}{\lambda} = R_H \times 1^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$ ...(i)

$\frac{1}{\lambda_{He^+}} = R_H \times 2^2 \times \left(\frac{1}{4} - \frac{1}{9}\right)$ ...(ii)

From (i) and (ii) $\frac{\lambda_{He^+}}{\lambda} = \frac{9}{5}$

$$\lambda_{He^+} = \lambda \times \frac{9}{5}, \Rightarrow \lambda_{He^+} = \frac{9\lambda}{5}$$

79. For first line of Paschen series n = 4 and for the second line n = 5

$$\frac{1}{(\lambda_1)_p} = R_H Z^2\left(\frac{1}{9} - \frac{1}{16}\right)$$

$$\frac{1}{(\lambda_2)_p} = R_H Z^2\left(\frac{1}{9}-\frac{1}{25}\right)$$

$$\frac{(\lambda_2)_p}{(\lambda_1)_p} = \frac{\frac{7}{16\times 9}}{\frac{16}{25\times 9}} = \frac{25\times 7}{16\times 16}$$

$$(\lambda_2)_p = \frac{25\times 7}{16\times 16}\times 720,\ (\lambda_2)_p = 492 \text{ nm}$$

kya ho raha hai, dimag ghum gaya pura

81. 10.2 eV energy is required to excite an electron from—

n = 1 to n = 2 in H-atom sample

n = 2 to n = 4 in He^+ ion sample

n = 3 to n = 6 in Li^{+2} sample

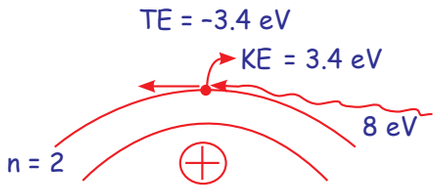
Now total maximum photons emitted when electrons jumps from these upper level to ground level are—

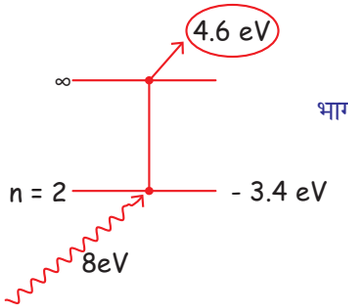
$$= 1 + \frac{4(4-1)}{2} + \frac{6(6-1)}{2} = 22$$

But unique photon = 22 - 2 = 20 because $2 \rightarrow 1$ transition in H-atom is equi energy with $4 \rightarrow 2$ in He^+ and $6 \rightarrow 3$ in Li^{+2}

85. $$= \frac{\frac{30}{3}Å - \frac{70}{9}Å}{\frac{20}{3}Å} = \frac{1}{3}$$

87.





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$$\left(\lambda = \sqrt{\frac{150}{4.6}}A^\circ\right)$$

89. First calculate increase in kinetic energy due to acceleration:

qV = e × 2V = 2eV

Final K E = 4 eV

$$\text{Now, } \lambda = \frac{12.27}{\sqrt{4}} = 6.15Å$$

90. $\lambda_1 = \sqrt{\frac{150}{100}}, \lambda_2 = \sqrt{\frac{150}{8.1}}, \lambda_3 = \sqrt{\frac{150}{49}}$

$$\frac{\lambda_3-\lambda_2}{\lambda_1} = \frac{\sqrt{\frac{150}{49}}-\sqrt{\frac{150}{81}}}{\sqrt{\frac{150}{100}}} = \frac{\frac{1}{49}-\frac{1}{81}}{\frac{1}{100}} = \frac{\frac{1}{7}-\frac{1}{9}}{\frac{1}{10}} = \frac{20}{63}$$

thoda toh hai bhayi muskil i also accept

92. Final kinetic Energy (KE_2) = KE_1 + eV

= 6eV + 4eV = 10eV

$KE_2 = p^2/(2m)$

$p = \sqrt{2m\times KE_2}$

$= \sqrt{2\times 9.109\times 10^{-31}\text{kg}\times 10\text{eV}\times 1.602\times 10^{-19}\text{J/eV}}$

= 1.707 × 10^{-24}kg m/s

λ = h/p

λ = 6.626× 10^{-34} J s / 1.707 × 10^{-24}kg m/s

= 3.883 × 10^{-10} m = 0.3883 nm

darne ka nahi, bas formula hi lgaya hai

100. $\Delta x\cdot\Delta v \geq \frac{h}{4\pi m},\ \Delta x \geq \frac{h}{4\pi m\Delta v}$

$$\geq \frac{0.53\times 10^{-34}}{\Delta v},\ \Delta v = \frac{45\times 2}{100}\text{m/sec} = 0.9 \text{ m/s}$$

109. [16] For n = 4 we have

	4s	4p	4d	4f
Total e	2	6	10	14
Total no. of electron with "S" = + $\frac{1}{2}$	1	3	5	7

oye balle balle, shava shava

115. [6] n = 4

ℓ	m_l
0	0
1	– 1, 0, +1
2	– 2, – 1, 0, +1, +2
3	–3, –2, –1, 0, 1, 2,3

So, number of orbital associated with

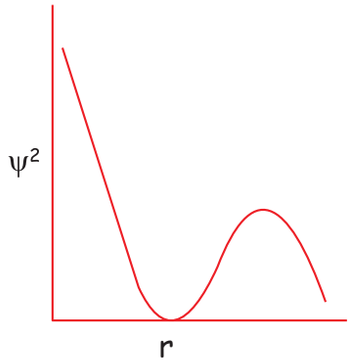
n = 4, $[m_\ell]$ = 1 are 6

Now, each orbital contain one e^- with

$m_s = -\frac{1}{2}$

Hence, total number of electrons with the above quantum number are 6.

119. [3] 1, 4 and 5 statements are correct.



For 2s orbital, the probability density first decreases and then increases. Angular nodes for p and d-orbitals is 1 and 2. At any distance from nucleus the probability density of finding electron is never zero and it always have some finite value.

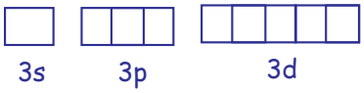
padho padho, sawaal karo

120. (B) For a given shell, the total number of allowed orbital is given by n^2

Consider M – shell, n = 3

So, number of allowed orbital will be 9.

It can be seen as,



So, statement-I is correct.

For any subshell, the spatial or magnetic quantum number is given by $2\ell + 1$ or $-\ell$ to $+\ell$ values including zero.

So, statement-II is correct.

121. (D) For H-atoms, orbitals in the shell are degenerate, meaning energy depends only on the value of 'n'.

$\therefore E_{3p_x} = E_{3d_{x^2-y^2}} = E_{3d_{z^2}}$

122. (D) The energy of an electron in a hydrogen atom is determined solely by the principal quantum number. Thus, the energy of the orbitals in hydrogen atom increases as follows :

1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f